IBS PRECISION ENGINEERING

Research & Development Department | Eindhoven, The Netherlands

Project Technical Brief & Scope Report

S7 Engineering Project: "Find the Force"

Development of an Opto-Mechanical Micro-Force Sensing System

Date: October 2026
Classification: Technical Assessment

1 Executive Summary & Academic Overview

This technical document outlines the engineering scope and multidisciplinary deliverables of the **"Find the Force"** project, developed in direct collaboration with **IBS Precision Engineering** (Eindhoven). The initiative addresses a critical high-precision industry challenge: measuring sub-Newton axial forces ($0 - 1\,N$) in restricted vacuum environments on high-speed vacuum nozzles without using physical signal cables.

This report serves as an explicit summary for the academic evaluation committee, highlighting the project's direct integration of three fundamental engineering competencies:

- **Materials Engineering:** Characterization and processing of non-silicone thermoplastic elastomers (SEBS) with strict mechanical constraints ($E = 1.15\,MPa$, $\nu = 0.48$, hysteresis $< 1\%$) to eliminate outgassing and out-of-plane buckling under load.
- **Manufacturing Engineering:** Development of direct micro-grating fabrication workflows and the *complete mechanical and mechatronic engineering of a custom laser-based force calibration machine*.
- **Production:** Transitioning from single-prototype molding to a scalable, repeatable batch manufacturing process for industrial nozzle integration.

2 Material Science & Engineering Constraints

The sensor relies on an elastic polymer element featuring a micro-scale diffraction grating. Force applied compresses the polymer, altering grating pitch and shifting a reflected $405\,nm$ laser beam. Material selection and mechanical characterization are central to project success:

- **Target Mechanical Properties:** The elastic body utilizes a specialized **SEBS** (Styrene-Ethylene-Butylene-Styrene) formulation engineered to achieve a Young's modulus (E) of

1.15 MPa, a Poisson's ratio (ν) of 0.48, and a mechanical hysteresis of less than 1%. These parameters guarantee linear response and eliminate optical drift under cyclic loading.

- **Strict Silicone Elimination:** Silicone polymers are strictly prohibited due to outgassing risks in vacuum environments, which contaminate high-precision optical lenses and degrade long-term structural integrity.
- **Structural & Deformation Mechanics:** Detailed non-linear analysis is conducted to eliminate out-of-plane bowing and buckling under axial loads, ensuring the micro-grating surface remains planar during compression.

3 Manufacturing Engineering & Custom Test Rig Development

To overcome the 11% measurement deviation of early prototypes, the engineering team is required to overhaul the manufacturing process and build an integrated experimental setup from the ground up:

- **Direct Micro-Grating Manufacturing:** Eliminating multi-stage indirect molding (3D master $\rightarrow$ silicone mold $\rightarrow$ part) in favor of direct micro-imprinting or high-resolution additive methods on SEBS to preserve micro-grating resolution.
- **Design & Construction of Custom Laser Test Rig:** The team is responsible for the full mechanical design, structural assembly, sensor integration, and programming of a dedicated laser-measurement testing machine. Key development phases include:
  - **Optical & Sensor Setup:** Selecting and mounting 405 nm diode laser sources, precision beam-splitting optics, and optical position-sensitive detectors (PSD/CMOS).
  - **Signal & Frequency Tuning:** Configuring optical sampling frequencies and sensor response thresholds to capture micro-shifts in diffracted laser beams.
  - **System Calibration & Testing:** Designing low-friction mechanical guide stages, calibrating axial force application using reference load cells, and running automated experimental validation protocols.

4 Production & Scalability

Industrial implementation by IBS Precision Engineering requires outfitting multiple vacuum nozzle assemblies with uniform sensor units:

- **Cycle Time & Process Optimization:** Replacing indirect casting workflows to streamline production throughput and eliminate cumulative tolerance errors across intermediate manufacturing steps.
- **Batch Quality Control & Repeatability:** Establishing standardized fabrication protocols, quality assurance metrics, and tolerance stack-up models to guarantee sub-Newton precision across serial production runs.

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